

Time and Work - Formulas

Work from Days:

If A can do a piece of work in n days, then A's 1 day's work = $\frac{1}{n}$.
--

Days from Work:

If A's 1 day's work = $\frac{1}{n}$,	then A can finish the work in n days.
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Ratio:

If A is thrice as good a workman as B, then:

Ratio of work done by A and B = 3 : 1.

Ratio of times taken by A and B to finish a work = 1 : 3.

Time and Work - General Questions

1. A can do a work in 15 days and B in 20 days. If they work on it together for 4 days, then the fraction of the work that is left is :

1
4

1
10

7
15

8
15

2. A can lay railway track between two given stations in 16 days and B can do the same job in 12 days. With help of C, they did the job in 4 days only. Then, C alone can do the job in:

9	$\frac{1}{5}$	days
---	---------------	------

9	$\frac{2}{5}$	days
---	---------------	------

9	$\frac{3}{5}$	days
---	---------------	------

10 days

3. A, B and C can do a piece of work in 20, 30 and 60 days respectively. In how many days can A do the work if he is assisted by B and C on every third day?

12 days

15 days

16 days

18 days

4. A is thrice as good as workman as B and therefore is able to finish a job in 60 days less than B. Working together, they can do it in:

20 days

22	1	days
	2	

25 days

30 days

5. A alone can do a piece of work in 6 days and B alone in 8 days. A and B undertook to do it for Rs. 3200. With the help of C, they completed the work in 3 days. How much is to be paid to C?

Rs. 375

Rs. 400

Rs. 600

Rs. 800

6. If 6 men and 8 boys can do a piece of work in 10 days while 26 men and 48 boys can do the same in 2 days, the time taken by 15 men and 20 boys in doing the same type of work will be:

4 days

5 days

6 days

7 days

7. A can do a piece of work in 4 hours; B and C together can do it in 3 hours, while A and C together can do it in 2 hours. How long will B alone take to do it?

8 hours

10 hours

12 hours

24 hours

8. A can do a certain work in the same time in which B and C together can do it. If A and B together could do it in 10 days and C alone in 50 days, then B alone could do it in:

15 days

20 days

25 days

30 days

9. A does 80% of a work in 20 days. He then calls in B and they together finish the remaining work in 3 days. How long B alone would take to do the whole work?

23 days

37 days

$37\frac{1}{2}$

40 days

10. A machine P can print one lakh books in 8 hours, machine Q can print the same number of books in 10 hours while machine R can print them in 12 hours. All the machines are started at 9 A.M. while machine P is closed at 11 A.M. and the remaining two machines complete work. Approximately at what time will the work (to print one lakh books) be finished ?

11:30 A.M.

12 noon

12:30 P.M.

1:00 P.M.

11. A can finish a work in 18 days and B can do the same work in 15 days. B worked for 10 days and left the job. In how many days, A alone can finish the remaining work?

5

5	1
	2

6

8

12. 4 men and 6 women can complete a work in 8 days, while 3 men and 7 women can complete it in 10 days. In how many days will 10 women complete it?

35

40

45

50

13. A and B can together finish a work 30 days. They worked together for 20 days and then B left. After another 20 days, A finished the remaining work. In how many days A alone can finish the work?

40

50

54

60

14. P can complete a work in 12 days working 8 hours a day. Q can complete the same work in 8 days working 10 hours a day. If both P and Q work together, working 8 hours a day, in how many days can they complete the work?

5	5
	11

5	6
	11

6	5
	11

6	6
	11

15. 10 women can complete a work in 7 days and 10 children take 14 days to complete the work. How many days will 5 women and 10 children take to complete the work?

3

5

7

Cannot be determined

None of these

16. X and Y can do a piece of work in 20 days and 12 days respectively. X started the work alone and then after 4 days Y joined him till the completion of the work. How long did the work last?

6 days

10 days

15 days

20 days

17. A is 30% more efficient than B. How much time will they, working together, take to complete a job which A alone could have done in 23 days?

11 days

13 days

20	3	days
	17	

None of these

18. Ravi and Kumar are working on an assignment. Ravi takes 6 hours to type 32 pages on a computer, while Kumar takes 5 hours to type 40 pages. How much time will they take, working together on two different computers to type an assignment of 110 pages?

7 hours 30 minutes

8 hours

8 hours 15 minutes

8 hours 25 minutes

19. A, B and C can complete a piece of work in 24, 6 and 12 days respectively. Working together, they will complete the same work in:

1	day
24	

7	day
24	

3	3	days
	7	

4 days

20. Sakshi can do a piece of work in 20 days. Tanya is 25% more efficient than Sakshi. The number of days taken by Tanya to do the same piece of work is:

15

16

18

25

21. A takes twice as much time as B or thrice as much time as C to finish a piece of work. Working together, they can finish the work in 2 days. B can do the work alone in:

4 days

6 days

8 days

12 days

22. A and B can complete a work in 15 days and 10 days respectively. They started doing the work together but after 2 days B had to leave and A alone completed the remaining work. The whole work was completed in :

8 days

10 days

12 days

15 days

23. A and B can do a piece of work in 30 days, while B and C can do the same work in 24 days and C and A in 20 days. They all work together for 10 days when B and C leave. How many days more will A take to finish the work?

18 days

24 days

30 days

36 days

24. A works twice as fast as B. If B can complete a work in 12 days independently, the number of days in which A and B can together finish the work in :

4 days

6 days

8 days

18 days

25. Twenty women can do a work in sixteen days. Sixteen men can complete the same work in fifteen days. What is the ratio between the capacity of a man and a woman?

3 : 4

4 : 3

5 : 3

Data inadequate

26. A and B can do a work in 8 days, B and C can do the same work in 12 days. A, B and C together can finish it in 6 days. A and C together will do it in :

4 days

6 days

8 days

12 days

27. A can finish a work in 24 days, B in 9 days and C in 12 days. B and C start the work but are forced to leave after 3 days. The remaining work was done by A in:

5 days

6 days

10 days

10	1	days
	2	

28. X can do a piece of work in 40 days. He works at it for 8 days and then Y finished it in 16 days. How long will they together take to complete the work?

13	1	days
	3	

15 days

20 days

26 days

29. A and B can do a job together in 7 days. A is $1\frac{3}{4}$ times as efficient as B. The same job can be done by A alone in :

9	1	days
	3	

11 days

12	1 4	days
----	---------------------------	------

16	1 3	days
----	---------------------------	------

30. A and B together can do a piece of work in 30 days. A having worked for 16 days, B finishes the remaining work alone in 44 days. In how many days shall B finish the whole work alone?

30 days

40 days

60 days

70 days

Time and Work - Data Sufficiency 1

Directions to Solve

Each of the questions given below consists of a statement and / or a question and two statements numbered I and II given below it. You have to decide whether the data provided in the statement(s) is / are sufficient to answer the given question. Read the both statements and

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- Give answer (D) if the data even in both Statements I and II together are not sufficient to answer the question.
- Give answer (E) if the data in both Statements I and II together are necessary to answer the question.

1.

A and B together can complete a task in 7 days. B alone can do it in 20 days. What part of the work was carried out by A?

I. A completed the job alone after A and B worked together for 5

II Part of the work done by A could have been done by B and C

I alone sufficient while II alone not sufficient to answer

II alone sufficient while I alone not sufficient to answer

Either I or II alone sufficient to answer

Both I and II are not sufficient to answer

Both I and II are necessary to answer

2.

How long will Machine Y, working alone, take to produce x	
I.	Machine X produces x candles in 5 minutes.
II	Machine X and Machine Y working at the same time produce x

I alone sufficient while II alone not sufficient to answer

II alone sufficient while I alone not sufficient to answer

Either I or II alone sufficient to answer

Both I and II are not sufficient to answer

Both I and II are necessary to answer

Time and Work - Data Sufficiency 2

Directions to Solve

Each of the questions given below consists of a question followed by three statements. You have to study the question and the statements and decide which of the statement(s) is/are necessary to answer the question.

1.

In how many days can 10 women finish a work?	
I.	10 men can complete the work in 6 days.
II.	10 men and 10 women together can complete the work in 3 3 days 7
III	If 10 men work for 3 days and thereafter 10 women replace . them, the remaining work is completed in 4 days.

Any two of the three

I and II only

II and III only

I and III only

None of these

2.

How many workers are required for completing the construction	
I.	20% of the work can be completed by 8 workers in 8 days.
II.	20 workers can complete the work in 16 days.
III.	One-eighth of the work can be completed by 8 workers in 5

I only

II and III only

III only

I and III only

Any one of the three

Simple Interest - Formulas

1. Principal:

The money borrowed or lent out for a certain period is called the principal or the sum.

2. Interest:

Extra money paid for using other's money is called interest.

3. Simple Interest (S.I.):

If the interest on a sum borrowed for certain period is reckoned uniformly, then it is called simple interest.

Let Principal = P, Rate = R% per annum (p.a.) and Time = T years. Then

$$(i). \text{ Simple Interest} = \left(\frac{P \times R \times T}{100} \right)$$

$$(ii). P = \left(\frac{100 \times \text{S.I.}}{R \times T} \right) ; R = \left(\frac{100 \times \text{S.I.}}{P \times T} \right) \text{ and } T = \left(\frac{100 \times \text{S.I.}}{P \times R} \right).$$

Simple Interest - General Questions

1. A sum of money at simple interest amounts to Rs. 815 in 3 years and to Rs. 854 in 4 years. The sum is:

Rs. 650

Rs. 690

Rs. 698

Rs. 700

2. Mr. Thomas invested an amount of Rs. 13,900 divided in two different schemes A and B at the simple interest rate of 14% p.a. and 11% p.a. respectively. If the total amount of simple interest earned in 2 years be Rs. 3508, what was the amount invested in Scheme B?

Rs. 6400

Rs. 6500

Rs. 7200

Rs. 7500

None of these

3. A sum fetched a total simple interest of Rs. 4016.25 at the rate of 9 p.c.p.a. in 5 years. What is the sum?

Rs. 4462.50

Rs. 8032.50

Rs. 8900

Rs. 8925

None of these

4. How much time will it take for an amount of Rs. 450 to yield Rs. 81 as interest at 4.5% per annum of simple interest?

3.5 years

4 years

4.5 years

5 years

5. Reena took a loan of Rs. 1200 with simple interest for as many years as the rate of interest. If she paid Rs. 432 as interest at the end of the loan period, what was the rate of interest?

3.6

6

18

Cannot be determined

None of these

6. A sum of Rs. 12,500 amounts to Rs. 15,500 in 4 years at the rate of simple interest. What is the rate of interest?

3%

4%

5%

6%

None of these

7. An automobile financier claims to be lending money at simple interest, but he includes the interest every six months for calculating the principal. If he is charging an interest of 10%, the effective rate of interest becomes:

10%

10.25%

10.5%

None of these

8. A lent Rs. 5000 to B for 2 years and Rs. 3000 to C for 4 years on simple interest at the same rate of interest and received Rs. 2200 in all from both of them as interest. The rate of interest per annum is:

5%

7%

7	1	%
	8	

10%

9. A sum of Rs. 725 is lent in the beginning of a year at a certain rate of interest. After 8 months, a sum of Rs. 362.50 more is lent but at the rate twice the former. At the end of the year, Rs. 33.50 is

earned as interest from both the loans. What was the original rate of interest?

3.6%

4.5%

5%

6%

None of these

10. A man took loan from a bank at the rate of 12% p.a. simple interest. After 3 years he had to pay Rs. 5400 interest only for the period. The principal amount borrowed by him was:

Rs. 2000

Rs. 10,000

Rs. 15,000

Rs. 20,000

11. A sum of money amounts to Rs. 9800 after 5 years and Rs. 12005 after 8 years at the same rate of simple interest. The rate of interest per annum is:

5%

8%

12%

15%

12. What will be the ratio of simple interest earned by certain amount at the same rate of interest for 6 years and that for 9 years?

1 : 3

1 : 4

2 : 3

Data inadequate

None of these

13. A certain amount earns simple interest of Rs. 1750 after 7 years. Had the interest been 2% more, how much more interest would it have earned?

Rs. 35

Rs. 245

Rs. 350

Cannot be determined

None of these

14. A person borrows Rs. 5000 for 2 years at 4% p.a. simple interest. He immediately lends it to another person at $6\frac{1}{4}\%$ p.a for 2 years. Find his gain in the transaction per year.

Rs. 112.50

Rs. 125

Rs. 225

Rs. 167.50

Simple Interest - Data Sufficiency 1

Directions to Solve

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1.

The simple interest on a sum of money is Rs. 50. What is the sum?	
I.	The interest rate is 10% p.a.
II.	The sum earned simple interest in 10 years.

I alone sufficient while II alone not sufficient to answer

II alone sufficient while I alone not sufficient to answer

Either I or II alone sufficient to answer

Both I and II are not sufficient to answer

Both I and II are necessary to answer

2.

What is the sum which earned interest?	
I.	The total simple interest was Rs. 7000 after 7 years.
II.	The total of sum and simple interest was double of the sum after

I alone sufficient while II alone not sufficient to answer

II alone sufficient while I alone not sufficient to answer

Either I or II alone sufficient to answer

Both I and II are not sufficient to answer

Both I and II are necessary to answer

3.

What percentage of simple interest per annum did Anand pay to	
I	Anand borrowed Rs. 8000 from Deepak for four years.
II	Anand returned Rs. 8800 to Deepak at the end of two years and

I alone sufficient while II alone not sufficient to answer

II alone sufficient while I alone not sufficient to answer

Either I or II alone sufficient to answer

Both I and II are not sufficient to answer

Both I and II are necessary to answer

4.

What is the rate of simple interest?	
I.	The total interest earned was Rs. 4000.
II.	The sum was invested for 4 years.

I alone sufficient while II alone not sufficient to answer

II alone sufficient while I alone not sufficient to answer

Either I or II alone sufficient to answer

Both I and II are not sufficient to answer

Both I and II are necessary to answer

Simple Interest - Data Sufficiency 2

Directions to Solve

Each of the questions given below consists of a question followed by three statements. You have to study the question and the statements and decide which of the statement(s) is/are necessary to answer the question.

1.

What is the principal sum?	
I.	The sum amounts to Rs. 690 in 3 years at S.I.
II.	The sum amounts to Rs. 750 in 5 years at S.I.
III.	The rate of interest is 5% p.a.

I and III only

II and III only

I and II only

I and III only, or II and III only

Any two of the three

Compound Interest - Formulas

1. Let Principal = P, Rate = R% per annum, Time = n years.

2. When interest is compound Annually:

$$\text{Amount} = P \left(1 + \frac{R}{100} \right)^n$$

3. When interest is compounded Half-yearly:

$$\text{Amount} = P \left[1 + \frac{(R/2)}{100} \right]^{2n}$$

4. When interest is compounded Quarterly:

$$\text{Amount} = P \left[1 + \frac{(R/4)}{100} \right]^{4n}$$

5. When interest is compounded Annually but time is in fraction, say $3\frac{2}{5}$ years.

$$\text{Amount} = P \left(1 + \frac{R}{100} \right)^3 \times \left(1 + \frac{\frac{2}{5}R}{100} \right)$$

6. When Rates are different for different years, say $R_1\%$, $R_2\%$, $R_3\%$ for 1st, 2nd and 3rd year respectively.

$$\text{Then, Amount} = P \left(1 + \frac{R_1}{100} \right) \left(1 + \frac{R_2}{100} \right) \left(1 + \frac{R_3}{100} \right).$$

7. Present worth of Rs. x due n years hence is given by:

$$\text{Present Worth} = \frac{x}{\left(1 + \frac{R}{100} \right)^n}$$

Compound Interest - General Questions

1. A bank offers 5% compound interest calculated on half-yearly basis. A customer deposits Rs. 1600 each on 1st January and 1st July of a year. At the end of the year, the amount he would have gained by way of interest is:

Rs. 120

Rs. 121

Rs. 122

Rs. 123

2. The difference between simple and compound interests compounded annually on a certain sum of money for 2 years at 4% per annum is Re. 1. The sum (in Rs.) is:

625

630

640

650

3. There is 60% increase in an amount in 6 years at simple interest. What will be the compound interest of Rs. 12,000 after 3 years at the same rate?

Rs. 2160

Rs. 3120

Rs. 3972

Rs. 6240

None of these

4. What is the difference between the compound interests on Rs. 5000 for $1\frac{1}{2}$ years at 4% per annum compounded yearly and half-yearly?

Rs. 2.04

Rs. 3.06

Rs. 4.80

Rs. 8.30

5. The compound interest on Rs. 30,000 at 7% per annum is Rs. 4347. The period (in years) is:

2

2	1
	2

3

4

6. What will be the compound interest on a sum of Rs. 25,000 after 3 years at the rate of 12 p.c.p.a.?

Rs. 9000.30

Rs. 9720

Rs. 10123.20

Rs. 10483.20

None of these

7. At what rate of compound interest per annum will a sum of Rs. 1200 become Rs. 1348.32 in 2 years?

6%

6.5%

7%

7.5%

8. The least number of complete years in which a sum of money put out at 20% compound interest will be more than doubled is:

3

4

5

6

9. Albert invested an amount of Rs. 8000 in a fixed deposit scheme for 2 years at compound interest rate 5 p.c.p.a. How much amount will Albert get on maturity of the fixed deposit?

Rs. 8600

Rs. 8620

Rs. 8820

None of these

10. The effective annual rate of interest corresponding to a nominal rate of 6% per annum payable half-yearly is:

6.06%

6.07%

6.08%

6.09%

11. Simple interest on a certain sum of money for 3 years at 8% per annum is half the compound interest on Rs. 4000 for 2 years at 10% per annum. The sum placed on simple interest is:

Rs. 1550

Rs. 1650

Rs. 1750

Rs. 2000

12. If the simple interest on a sum of money for 2 years at 5% per annum is Rs. 50, what is the compound interest on the same at the same rate and for the same time?

Rs. 51.25

Rs. 52

Rs. 54.25

Rs. 60

13. The difference between simple interest and compound on Rs. 1200 for one year at 10% per annum reckoned half-yearly is:

Rs. 2.50

Rs. 3

Rs. 3.75

Rs. 4

None of these

14. The difference between compound interest and simple interest on an amount of Rs. 15,000 for 2 years is Rs. 96. What is the rate of interest per annum?

8

10

12

Cannot be determined

None of these

15. The compound interest on a certain sum for 2 years at 10% per annum is Rs. 525. The simple interest on the same sum for double the time at half the rate percent per annum is:

Rs. 400

Rs. 500

Rs. 600

Rs. 800

Compound Interest - Data Sufficiency 1

Directions to Solve

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1.

What is the rate of compound interest?	
I.	The principal was invested for 4 years.
II.	The earned interest was Rs. 1491.

I alone sufficient while II alone not sufficient to answer

II alone sufficient while I alone not sufficient to answer

Either I or II alone sufficient to answer

Both I and II are not sufficient to answer

Both I and II are necessary to answer

2.

What will be compounded amount?

I. Rs. 200 was borrowed for 192 months at 6% compounded

II. Rs. 200 was borrowed for 16 years at 6%.
--

I alone sufficient while II alone not sufficient to answer

II alone sufficient while I alone not sufficient to answer

Either I or II alone sufficient to answer

Both I and II are not sufficient to answer

Both I and II are necessary to answer

3.

An amount of money was lent for 3 years. What will be the difference between the simple and the compound interest earned
--

I. The rate of interest was 8 p.c.p.a.
--

II. The total amount of simple interest was Rs. 1200.

I alone sufficient while II alone not sufficient to answer

II alone sufficient while I alone not sufficient to answer

Either I or II alone sufficient to answer

Both I and II are not sufficient to answer

Both I and II are necessary to answer

Compound Interest - Data Sufficiency 2

Directions to Solve

Each of the questions given below consists of a question followed by three statements. You have to study the question and the statements and decide which of the statement(s) is/are necessary to answer the question.

1.

What is the rate of interest p.c.p.a.?	
I.	An amount doubles itself in 5 years on simple interest.
II.	Difference between the compound interest and the simple interest earned on a certain amount in 2 years is Rs. 400.
III.	Simple interest earned per annum is Rs. 2000.
.	

I only

II and III only

All I, II and III

Any two of the three

I only or II and III only

2.

What will be the compound interest earned on an amount of Rs.	
I.	The simple interest on the same amount at the same rate of
II.	The compound interest and the simple interest earned in one
III.	The amount becomed more than double on compound interest

I only

I and II only

II and III only

I and III only

None of these

Compound Interest - Data Sufficiency 3

Directions to Solve

In each of the following questions, a question is asked and is followed by three statements. While answering the question, you may or may not require the data provided in all the statements. You have to read the question and the three statements and then decide whether the question can be answered with any one or two of the statements or all the three statements are required to answer the question. The answer number bearing the statements, which can be dispensed with, if any, while answering the question is your answer.

1.

Mr. Gupta borrowed a sum of money on compound interest. What will be the amount to be repaid if he is repaying the entire amount

- | | |
|------|---|
| I. | The rate of interest is 5 p.c.p.a. |
| II. | Simple interest fetched on the same amount in one year is Rs. |
| III. | The amount borrowed is 10 times the simple interest in 2 years. |

I only

III only

I or II only

II and Either I or III only

All I, II and III are required

2.

What is the compound interest earned at the end of 3 years?

- | | |
|-----|---|
| I. | Simple interest earned on that amount at the same rate and for the same period is Rs. 4500. |
| II. | The rate of interest is 10 p.c.p.a. |
| III | Compound interest for 3 years is more than the simple interest |

I and II only

II and III only

I and III only

I and Either II or III only

Any two of the three